

Owen Thomas

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Profile

MSc Theoretical Physics student with a focus on quantum computing and information, currently working on topological quantum codes for my master's thesis. Proficient in Python (data analysis, hardware control and automation), quantum computation (quantum information theory, Qiskit for algorithm and circuit design, familiar with TKET) and ML (TensorFlow), with a proven ability in independent research. Interested in QEC-aware and ML based quantum circuit optimisation.

Education

King's College London, MSc Theoretical Physics Sept 2024 - Sept 2026 (part-time)

University of Exeter, BSc (Hons) Physics Sept 2021 – May 2024

Professional Experience

Research Intern - Experimental Optics, University of Exeter (CMRI) June 2024 – July 2024

- Designed an effective modular architecture for construction of spectral hypercubes, configuring super-continuum lasers, tunable filters, spectrometers, cameras and flexible beam-directors on an optical workbench.
- Integrated SDKs for spectral hardware into well documented and easy to use code, enabling tailored wavelength sweeps and data collection.
- Developed an intuitive Python-based GUI to automate hardware experiments and visualise data for hyperspectral imaging research.
- Worked independently to address and overcome challenging problems and meet key milestones, verifying progress with regular meetings.

Mathematics RLHF AI Training, Outlier AI January 2024 – June 2024

- Crafted original, graduate level mathematical prompts to challenge an LLM and force reasoning errors.
- Evaluated and refined LLM performance with high accuracy to improve mathematical reasoning via direct human feedback, taking advantage of the RLHF optimisation technique.

Academic Tutor, Self-employed 2021 - 2024

- Provided academic tutoring to As and A-level students in Physics, Maths and Further Maths, working with each students respective strengths to deliver personalised teaching.
- Fostered a positive and encouraging environment that allowed students to explore their perceived mathematical weak-points, observing tangible improvements in their confidence and mathematical ability.

Technical Skills

Programming:

- Python (TensorFlow, Pandas, NumPy, numerical simulation, hardware control and automation)
- QisKit (quantum circuit and algorithm development)
- Machine Learning (TensorFlow for deep learning, CNN/ PINNs for classification)

Scientific Analysis:

- Experimental Physics (independent research, literature review and analysis, experiment design and implementation, data analysis, modelling and simulation)
- Theoretical Physics (quantum mechanics, quantum computation and information, QFT, general relativity, linear algebra, advanced calculus, statistics)

Recent Projects

Programming: ML-based TKET quantum circuit optimisation, 5-Qubit stabiliser code, 9-Qubit Shor code, 3-Qubit repetition code for bit-flip protection

MSc Theoretical Physics: In the initial stages of my MSc research project on topological quantum codes, self-studying manifolds and fault-tolerant quantum computation whilst conducting preliminary literature search/ review.